

Karthick Sri Ram A

karthicksriram-portfolio.web.app | Software Developer

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EDUCATION

RAMCO INSTITUTE OF TECHNOLOGY

BE IN COMPUTER SCIENCE
2022 - 2026 | CGPA:8.37/10.0

CHINMAYA VIDYALAYA SLR MATRIC HR SEC SCHOOL

Grad. 2022 | Percentage: 89%

LINKS

Github:// karthick-320

LinkedIn:// karthicksriram

Leetcode:// karthicksriram_10

GFG:// karthicksriram10

SKILLS

Languages: Java, JavaScript, SQL

Frameworks/Libraries:

React.js, Node.js, Express.js, Tailwind CSS

Cloud & DevOps: AWS, Docker

AI & LLM: LangChain, RAG

Databases: MySQL, MongoDB, Neo4j, Redis

Tools: Git, GitHub, Postman, IntelliJ Idea, VS Code

Computer Science: Data Structure, Algorithms, OOPS, DBMS

COURSES

Programming in Java - NPTEL

Java Full Stack - Wipro talentNext

Full Stack Development - EMC Academy

EXPERIENCE

MAASR TECHNOLOGY | INTERNSHIP

Aug 2025 - Oct 2025 | Virtual

- Designed a normalized relational database schema (up to 3NF) supporting users, class groups, clubs, notes, and event registrations, minimizing 25% redundancy and maintaining query performance.
- Optimized database query performance by designing targeted single and composite indexes, reducing 75% retrieval latency for high-frequency operations.
- Integrated AWS S3 for storing user-uploaded files, enabling scalable and reliable file access.

PROJECTS

AI-POWERED DSA MOCK INTERVIEW PLATFORM |

REACT.JS, NODE.JS, EXPRESS.JS, MONGODB, REDIS, GPT LLM

Aug 2026 - Sep 2026

- Designed an AI workflow for DSA problem solving, generating automated follow-ups and evaluation feedback.
- Architected an asynchronous Docker-based execution engine using Redis and BullMQ, isolating workloads to support 500+ concurrent requests with sub-second API response latency.
- Secured REST APIs via Redis-backed rate limiting, rejecting 99.9% of excessive traffic during 1K+ request burst tests to prevent server overload.
- Optimized MongoDB queries with compound indexing, cutting average response time by 78.3% under load.

E-COMMERCE | REACT.JS, NODE.JS, EXPRESS.JS, MONGODB

Jun 2025 - Jul 2025

- Developed a Full-Stack marketplace using React.js and Express.js for students to trade regional snacks, utilizing MongoDB's flexible document-based storage.
- Designed a Dynamic Schema that allows different snack categories to have unique attributes.
- Ensured data integrity using ACID transactions for real-time inventory updates, preventing inconsistent states during concurrent operations.
- Implemented JWT-based authentication for secure user sessions.